Extra Problems

Please work on the problems below and then discuss the problems at the Week 3 Extra Problems Discussion forum.

Extra Problems week 3:

In this set, numbers can be, if you want, interpreted as sums of Hackenbush positions. Or just play the game until it's a number then stop and see who wins. The phrase "all numbers" means all p/q (or –p/q) with q a power of 2. As usual n.G is the sum of n copies of G and –n.G = n.(-G).

Games:

A = Up = {0| *}
B = {0 | -Up + *} = {0 | -A + *}
C = {0 | -A - B + *}
D = {0 | {0 | -2))
E = {0 | {0 | -100} and in general Tiny(G) = {0 | {0|-G}}
F = {1 | -1}
H= {5 | -5} and in general Check(G) = {G| -G}
I = run over ___ LR
J = run over ____ LR

1. Order A, B, C, D, E, -2,-1,0,1,2,3
2. Find all numbers x with E<x
3. Find all numbers y with y <E
4. Same as 2,3, but for E,F,H
5. Show that n.B < A
6. Show that n.C < B
7. Show that 2.D < A. Can you show n.D <C?
8. What the best strategy for playing Check(100) + Check(99) + Check(98) +...Check(1)?
9. Best strategy for playing Tiny(-3) + Tiny(1)
10. (Harder, I think) Who wins Tiny(Up) – Up?
11. Find all numbers y with y < I, and all x with I <x
12. Same for J.
13. Analyze 2.I, or in general 2n.I, with n a positive integer.
14. Same for 2.J and 2n.J.

That's All Folks